Lectures

Stat 151A: Linear Models

Lectures marked with a $*$ are still in progress.

These lecture notes are meant to supplement the reading indicated in the course schedule.

Course overview

These notes summarize the key topics in each unit.

Introduction

- Course philosophy
- Sample means
- Linear algebra review

Unit 1: Multlinear regression

- Simple linear regression
- Multilinear regression
- Paths to least squares
- Transforming the regressors
    - One-hot encodings
    - Interacting continuous regressors
    - Supplement: Linear transformations
- Transforming the response variable

Unit 2: Inference for linear regression

- Confidence intervals (review)
- Modeling of OLS
- OLS under normality
  - Supplement: The multivariate normal
  - Supplement: The distribution of the residual variance estimator
  - Variable selection and F-tests
- OLS under non-normality
  - Supplement: Asymptotics under misspecification
- Statistical versus practical significance

Unit 3: Machine learning and regularization

- OLS for prediction with expressive basis functions
- Bayesian methods and L2 regularization

Unit 4: How regression can go wrong

- The FWL theorem
- Regression to the mean
- Omitted variables
- Influence and outliers